

THE GRAVITATIONAL LENS AS AN INTERGALACTIC COMMUNICATION TOOL†

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Abstract—The gravitational lens, a consequence of general relativistic light bending with perfect or near-perfect observer–bender–object alignment, is now a well-studied natural phenomenon. Primarily an astrophysical tool for galactic and dark matter mass estimation, it may also have applications to SETI via gain enhancement of the ETI signal (a “gravity lens gain system”, or GLGS). It has been shown that gains of up to 80 dB may be achieved, making a GLGS a superior gain system for all save the most large-scale reflection systems. A “typical” application of a gravity lens may be realized through a special type of microlensing, where the transmitter (or receiver) orbits a compact or collapsed star. One or more satellites may sweep out a “lensing cone” which will result in an intense, transient signal at the observer sight. Unlike normal stars, collapsed stars afford very small ($\ll 1$ AU) orbits which allow for repetitive sweeps of the lensing cone across a region of sky. Steady-state alignment, although imaginable, may be impossible to achieve over interstellar or intergalactic distances. It is much more likely that the swept-gain orbiting system would be used in SETI. The gravitational lens may not be necessary for interstellar spans since sufficient EIRP can be achieved in other ways. However, it may be a method of choice for intergalactic transmissions since only Type III civilizations would be able to route sufficient EIRP's otherwise. Less advanced civilizations may use gravity lenses as a method of achieving very large EIRP's with their available technology. The signature of a lensed signal is a transient (less than 1 millisecond at peak) signal. For any degree of information to be conveyed the bandwidth must be large. Any inherent phase-frequency-amplitude modulation will be affected by the dynamics of the lens amplification and appropriate encoding/decoding schemes are discussed. High time resolution/broad bandwidth modes are suggested as a scenario for extragalactic SETI. VLBI is suggested for it allows for automatic bit checking of the extraterrestrial transmission.

1. INTRODUCTION

For over 10 years, the gravitational lens has been an established observational phenomenon (see, for example, [1]). Hypothesized long before its discovery, over sixty years went by before its acceptance as a natural, indeed inevitable, event. This appears surprising: the gravitational, or “gravity” lens requires no new physics. Indeed it is merely a specialized example of the gravitational deflection of light by a massive body, suggested by Einstein in 1936 as an example of general relativity in action.

That such a simple, magnifying system should exist was not to escape the attention of SETI researchers. Eshleman [2], Drake [3], and Cohen [4] have pointed out that the gravitational lens offers a marvelous advantage in providing gain in cosmic communication, and can also act as a bizarre cosmic telescope for the exploration of other worlds. Although such applications of gravity lensing are more science fiction than applied science, they are worth considering in ETI communication scenarios.

In this paper, we discuss the gravitational lens and its potential application as a tool in cosmic

communication, hereafter referring to it as a “gravity lens gain system” or GLGS. It is shown that under very restrictive conditions, any GLGS affords a viable means of assuring information transferral between galaxies. Specifically it opens up the opportunity for extragalactic SETI of an unusual type.

2. THE GRAVITATIONAL LENS

The warping of time and space by a gravitating mass causes a deflection of geodesic light paths, producing the gravitational refraction of light. Einstein [5] noted that this deflection causes more than a displacement of background stellar positions; perfect alignment between the light source (star), gravitating mass, and observer gives a magnified, remarkably distorted image (or multiple images) of the light source. Simple geometric considerations reveal that the point-like background source becomes ring-shaped (Einstein Ring) or, with slightly offset alignment, two or more non-point images. The gain in this lens effect is considerable. Einstein showed that the gain could approach infinity in the context of geometric optics. The lensing also enjoys the virtue of being achromatic and impartial to polarization of the light source.

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However, the gravitational refraction is normally so slight that only masses of a stellar mass or greater might be considered in the context of the effect, while the observer and light source must be at a considerable distance from the gravitating mass (the lens or "bender"). Similarly, the degree of alignment must be virtually perfect. Thus the likelihood of encountering the celestial superposition of two stars in this way was considered so slight as to make the issue of gravitational lenses purely theoretical.

However, extended masses of galaxies, and their vast numbers, made the likelihood of encountering this lens effect great. Indeed, since 1979, almost two dozen galactic gravity lenses have been found [1], all resulting from either perfect or near perfect alignment between distant galaxies from our vantage point.

The astrophysical importance of gravity lenses lies in the images. Shapes, numbers, and magnifications are all determined not only by the alignment, but by the mass and mass distribution of the bender. With the observed structures, limits to the bender's mass and mass distribution may be constrained. Especially important is the fact that the lens effect depends only on mass parameters of the bender, not its light output. Therefore gravity lenses can be used to ascertain dark matter quantities in galaxies.

Eshleman [2] viewed the lens effect in a SETI perspective. He found that diffraction effects limit the gain of lensing, at least with a star as a bender. Up to 80 dB of gain is afforded with perfect alignment; diffraction effects also introduce a slight frequency-dependent chromaticity. He cautioned on the practicality of gravity lenses as gain enhancers—perfect alignment requirements literally generate a spatial channel between communicating parties and eavesdropping from off the line of the two communicators will be impossible. The apparent cost of the high gain of a GLGS is limited spatial coverage.

3. HOW MIGHT GRAVITY LENSES BE USED?

Aware of the problems encountered with chance alignment of normal stars, Drake [3] and Cohen [4] explored other possibilities, aimed at using a GLGS to provide gain over greater sky coverage, which may be intercepted by many different civilizations at different celestial positions. Assuming the purposeful exploitation of the lensing effect, they found that exact alignment was difficult to maintain. For an orbiting transmitter (or receiver) the alignment could only be attained for short period of time, perhaps fractions of a second. Even at the minimum focal distance for a normal star, that is 550 AU, [2] the alignment is short lived. They considered the opposite extreme: could information be transferred if the satellite or orbiting object periodically entered the focal zone in GLGS? Repeating orbits demand very small orbital radii, otherwise the timescale for repetition may be very many orders of magnitude larger

than the timescale for alignment. With a minimum focal distance of 550 AU for a star like the Sun, there lies no hope for such repetitions. However, for collapsed stars, the minimum focal distance is quite small—0.4 solar radii for a white dwarf, a few kilometers for a neutron star. With such small orbital radii possible, the period of the repetitive "sweep", or the time between subsequent focal zone alignments, is given by

$$P = 2.1 \times 10^{-5} R^{1.5} \text{ s} \quad (1)$$

where R is the orbital radius in kilometers. For example, an orbital radius of 10,000 km yields a sweep-repetition of about 20 s. The signal, as received, would produce a comb function in time. During the "off" cycle, the lensing effect has not disappeared, rather it has swept past other sky directions, much like a lighthouse beacon. Thus orbiting satellites around collapsed stars could provide a platform for transmitting high gain, swept-mode transmissions through the GLGS or, alternatively, could be used to eavesdrop on other civilizations.

We have no hope of using a gravity lens in this manner. However, more advanced civilizations may employ this technique as an aid for other civilizations to detect their transmissions.

Where might we seek out the collapsed stars which would have sweeping satellites? To consider this for our own galaxy we would need to know where a high density of these special stars lies. Furthermore, Type II civilizations would probably be required to build and utilize the hardware necessary; Type I civilizations might exist in far greater numbers and would already have the means to transmit across the galaxy without the need of GLGS's through alternative and less exotic gain systems. Galactic SETI does not require gravity lenses to achieve the desired result.

The situation appears to be exacerbated for transmissions between galaxies. The great distances (> 10 Mpc) have discouraged their choice as observing targets. The concurrent dissipation of signal gains is in excess of 60 dB. Any hopes for receiving a transmission from another galaxy would require that a Type III, or at least an advanced Type II civilization, was responsible for the transmission. Although such gain requirements effectively act as a "high pass filter" for technological competence, they are aggravated by the likely rarity of Type III or Type II civilizations compared to presumably more numerous Type I civilizations. We might seek to pick up a signal from another galaxy, only to discover that no civilization with an adequately advanced technology exists within it.

However, with a GLGS, large gain factors are attained without the super advanced technology otherwise necessary. Thus less advanced civilizations, which may be more abundant in number, may produce EIRP's which rival those achieved by Type III civilizations with different and more difficult means to produce gain systems.

When a telescope has a galaxy within its beam, over 10^{10} stars are investigated at one time. If perhaps 1% of these stars are collapsed objects then each galaxy presents several tens of millions of ETI search targets simultaneously. If advanced Type I or modest Type II civilizations are present, then it may be likely that some have used collapsed stars as GLGS's. And since power dissipation from distance has been effectively overcome, dedicated extragalactic SETI becomes viable within the context of a GLGS. Indeed, since more objects are being targeted, this scenario may be more viable than targeted galactic searches of a small number of individual stars. Galaxies as a whole become excellent candidates for SETI, provided that orbited GLGS's exist in them.

4. COMMUNICATION MODES

Finding swept-beam GLGS's depends on much more than just the pointing of a telescope. The effectiveness of the communication mode is highly restricted by the lensing properties and orbital timescales. The signal could be misconstrued as a transient natural event, not unlike a stellar flare, if proper observing parameters are not chosen.

The outstanding characteristic of the ETI signal will be its short time span during a given orbital sweep through the focal zone. The signal will have a rapid rise and decay and resemble a Gaussian in time-intensity. Rise and decay times may be on the order of milliseconds. With such a fleeting signal in time, bit rates must be very high, kilobits per second or greater.

Corresponding bandwidths, as constrained by Gabor's relations [6] must be very large to convey information. These bandwidths may be octaves of megahertz. Indeed, the frequency profile of a GLGS transmission may not be unlike a spread-spectrum mode. Thus with a GLGS, the notion of a "magic frequency" is not critical, while a "magic frequency range" may be.

A principal distinction of a spatially unresolved GLGS signal is that it will manifest interference which will cause gain variations across the passband. The integrated signal, at a given instant, represents a phasor whose magnitude depends on the relative

magnification and time delays of the images. Since the phase difference is frequency dependent, a broadband signal will exhibit both constructive and destructive interference. The spectrum will actually change rapidly in time as the delays and intensities of the images change with the sweep through the focal zone.

It seems inevitable that arrays are required to detect GLGS's. First, since the signal is so fleeting, there is a need to corroborate the intelligent, celestial nature of the signal. In the absence of knowledge of the repetition time of the GLGS comb, each GLGS signal must be independently confirmed. It is the natural tendency to confirm phenomenon through an independent sampling in time. In fact, a simultaneous but independent detection also meets reproducibility requirements. Additionally, time of arrival differences with an array are key to discerning the farfield nature of the phenomenon and fix it in the celestial sphere.

An added benefit of array detection (VLBI) is that it makes it possible to resolve the lensed images during a comb sweep. During the rise and decay portions of the sweep, the image will break down into multiple crescents. With spatial resolution, each image appears offset in time from the others. Through cross-correlation it is possible to produce automatic bit checking in the encoded message, providing an essential redundancy for proper decoding. The GLGS may be totally non-redundantly encoded, in addition to its rapid bit rate.

We suggest that a viable scenario for SETI include extragalactic, broadband searches with rapid time resolution. VLBI could assist in decoding the message if received. Perhaps then we may test this most exotic of natural phenomenon in its unusual use by intelligent beings.

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